PANTONE REPORT : THE PAGES OF MICHEL DAVID

PAGE 17

IN-DEPTH REFLECTIONS ON THE PANTONE ASSEMBLY CALLED "Gillier-Pantone System"

Page created 03/11/2005

Back to the M.David menu

I intend this new page especially for those who, like me, find that the theory of things is a little boring and that one is better entertained by doing "practice". Unfortunately, if you don't have a few simplified theoretical bases in the field in which you work, you quickly find yourself cornered and, finally, you go around in circles and you waste your time instead of progressing.

In this page I will therefore endeavor to provide, somewhat out of order and by simplifying them to the extreme, certain explanations on the operation of engines and on various thermodynamic phenomena which concern them.
These explanations are essential for a refined understanding of my page 15 and especially on what I have overlooked in it.

I will explain below what makes the difference between a traditionally powered diesel engine and the same engine equipped with the Pantone water-only version that we have become accustomed to calling the “G system”.
It should immediately be noted that in the traditional version, the heat produced by the fuel is used only to heat the various gases present which, by their expansion, directly give us "work".

In the "system G" version, the fuel combustion is used rather to break down a more or less important quantity of very hot water vapor, previously admitted and compressed in the cylinder, by providing it with the missing calories in addition to those recovered. on the exhaust gases. It is the combustion or rather the explosion of the hydrogen and oxygen which have appeared which increases the temperature of the various gases present, most of which is water vapor, which by relaxing produces the work.

It should be noted that the decomposition of the greater or lesser part of water vapor (which depends on the quantity of fuel injected and the heat load of the vapor entering the cylinder), is done very cheaply since the load calorific value of this is to be deduced from the energy necessary for this decomposition.

This is where one of the causes of the low consumption of the engines thus equipped resides, but only one of the causes, because as I demonstrate below, the other cause is a significant improvement in the efficiency of these engines.

REMINDER ON THE OPERATION OF A CLASSIC ENGINE.

Before continuing, I think it is good to make a simplified reminder on the way in which the work is produced in an engine, that is to say on what really pushes the "piston" is what produces this famous work, thus, you will know a little more about it than those which are satisfied to say that it is the explosion which pushes this one. In fact, I will simply explain some relatively complex things to you which will appear obvious to you even if you are not familiar with thermodynamics.

So if you are angry with the theory, that is to say with thermodynamics and thermochemistry, at least remember this fact, that it is by the increase in volume of a gas existing in the cylinder between the piston and the cylinder head, or the appearance of a new gas and even both simultaneously, that the piston is "pushed". In fact, under such conditions it is understood that a gas can only increase in volume if it is able to push the piston, so that it can push the piston, its pressure must increase, so that its pressure increases, it is necessary to raise its temperature, for this it is necessary to bring it calories which is done in the engines by the combustion of a fuel. By burning, the fuel produces gases which are added to those present in addition to the heat energy, and that's good.

It is understood that the gas mentioned above is in the majority of cases carbon dioxide, water vapor and what remains of the air is nitrogen.

Meditate on this and with what follows you will have understood everything.

Let us return to the operation of our engine and take the case of "DIESEL".

Here is what is happening point by point:

After admission, at the end of compression, the intake air is compressed in the space between the piston and the cylinder head.
When combustion occurs at the start of the injection of fuel for the "diesel" the piston is pushed by the expansion of several different gaseous products which have increased in temperature and pressure thanks to three independent causes which start from the start of admission.

Thus in an engine at its operating temperature:

- 1- During admission, the outside air, assumed to be 20 °, heats up by entering the cylinder in contact with the intake valve of the cylinder walls and the cylinder head (it may be noted in passing that by heating up the air increases in volume and this ultimately ends with poor filling of the cylinder, ie an insufficient quantity of oxygen for high speeds).

During this operation, the air therefore increases in temperature by recovering calories and increases in volume for the same reason and in addition because it is in depression, which justifies the lack of oxygen necessary to burn the maximum amount of fuel. which is actually equivalent to a reduction in engine displacement. This is in addition to the inconvenience of admitting too hot outside air. To remedy this, we invented the compressor (turbo or other and air cooling systems etc.).

- 2- In the compression time, by going up, the piston compresses this air, which raises its temperature by volume reduction. But, at the same time, the temperature rise is not what it should be because some of the heat leaks into the mass of the engine in contact with the piston cylinder and the cylinder head, especially at the end of the stroke.

Despite everything, this air reaches a variable temperature depending on the type of engine and its speed. It is quite high in the case of the diesel engine since the compression ratio is important.

In this case, there is no recovery of calories on the walls of the cylinder, but a significant loss because of the temperature difference between the walls of the interior of the cylinder (less than 100 °) and the temperature of the cylinder. compressed air (much more than 100 °). This widens the gap between theoretical yield and practical yield.

NOTE: Until now, the engine has only consumed “work” already accumulated in the mass of moving engine parts and provided by the previous cycle.

- 3 - At the end of compression, the piston having reached top dead center, the volume of air already very hot sees its temperature increase considerably and suddenly by the combustion of the injected fuel (or already present for non-diesel engines). During this operation, the air gives up its oxygen to the fuel for combustion.
The gases resulting from this combustion are added to the nitrogen of the air, increasing the volume of gas (ie the pressure) which, by relaxing, pushes the piston.

Thus the different hot gases which, by expanding, push the piston towards bottom dead center, produce a quantity of work, part of which comes from the fuel and the other is a partial return of the work required by the compression and provided by the previous cycle. . It is obvious that the work provided will be cut off from what it takes to ensure the three cycles that will follow, that is to say the exhaust, the admission and the compression.

NATURE OF GAS BEFORE AND AFTER COMBUSTION.

It is interesting to examine closely the nature of the gaseous products which exist or arise during combustion.
NOTE: The combustion products of a hydrocarbon are mainly carbon dioxide and water vapor. As to burn the hydrocarbon (gasoline or fuel), we needed air which contains the necessary oxygen, so we also mixed with the products of combustion, the nitrogen of the air in question. If by chance there was

an excess of air we would end up with a little oxygen or carbon monoxide in case of lack.

To simplify, we can remember that the three main gaseous products which push the piston while expanding are:

- Carbon dioxide
- Water vapor
- The nitrogen of the air admitted at the start

I will answer immediately to those who are surprised to see water vapor in the exhaust gases of their engine and would like to know the cause.
Since hydrocarbons are mainly composed of carbon and hydrogen in variable proportions, it should be understood that part of the oxygen in the air will be the oxidizer of carbon and the rest will be the oxidizer of hydrogen.

Thus, the combustion of carbon with one part of oxygen gives carbon dioxide, and the combustion of hydrogen with the other part of oxygen gives water vapor which is often found condensed into water at the outlet. from an exhaust that is too cold.

In fact, in hydrocarbons we have two very different mixed fuels, which need oxygen to burn.

What those who "follow" will notice is that the presence of nitrogen can appear to be a disadvantage since it does not provide any energy, however, without its presence, an engine supplied only with pure oxygen to the gas. instead of air, would be more difficult to design mechanically because of the considerable temperatures which would be reached at the time of combustion.

While the presence of nitrogen is not a blessing, it does at least give us that benefit.

It is also necessary to consider that when the vapor is given off, at the time of the combustion of hydrogen, and joins the heated nitrogen and the carbon dioxide, the engine is thus partially a vapor engine.

Before I begin my chatting about the PANTONE System known as the “G-System”, I'm going to talk to you about performance and power.
In my discussions here and there with those interested in engines, I have found that many do not really see the difference between horsepower and efficiency.
Before going any further, I will try, simplifying it, to fix the problem once and for all.

So remember that an engine can be very powerful and have a poor performance and the opposite is also true. Also note that as a general rule, the more power is required from an engine, the more its efficiency decreases.

Efficiency and power are therefore two different things.

- Power is the amount of work that an engine is able to provide over a certain period of time **without considering the amount of energy used to obtain it.** By convention, when we say that an engine is 30 hp, that means that it can provide a force of 30 hp and maintain it over time. As a general rule, we mean "horse hour".

- Yield is what characterizes**the quantity of work obtained in relation to the quantity of energy committed to obtain it,** it is expressed as a percentage. We can say: this engine has an efficiency of 23%, if 23% of the gasoline consumed produced work.

That said, if the definition of power hardly deserves further explanation, it is more important to go into the definition of efficiency a bit.
It should be understood that there are several kinds of returns, including:

- The theoretical yield.
- The actual or practical (industrial) yield.
- Economic return.

The theoretical efficiency can be obtained by calculation, even before an engine is built. This can be determined on the basis of the mechanical characteristics, but mainly according to the compression ratio adopted and the type of engine, combustion or combustion (gasoline, gas or diesel).

The practical efficiency is approximately calculated by applying a reduction coefficient obtained empirically. It gives us a fairly exact proportion of the work expected in relation to the energy to be engaged. The reduction coefficient is known from experience and depends on the type of motor etc.
The practical yield obviously never reaches the theoretical yield, it is even generally very far from it.
The main causes which make it difficult for the practical efficiency to approach the theoretical efficiency are:

- The inevitable mechanical friction which can reach about 15%.
- The heat which escapes through the walls of the cylinder.
- Improper filling of the cylinder.
- The difficulty in evacuating the exhaust gases and especially the obligation of the truncated trigger which generally costs us more than 15% (see my page 16).
- The rate of ignition of the combustible or explosive mixture.

Economic return,is the same as the real efficiency since if the engine has an efficiency of 25%, in the price of the fuel expended, 25% will have produced work and the rest will be lost (except for the engines in fixed station where one sometimes recovers part of this loss for ancillary uses such as space heating).

NOTE: Even if you don't have a lot of engine knowledge, you have all heard that the “diesel” engine has exceptional performance, which justifies its low consumption compared to the gasoline engine.
This is true, this high efficiency which can approach 40% concerns the practical efficiency, because the theoretical efficiency of this type of engine is lower than the theoretical efficiency of the internal combustion engine (at the compression ratio supposedly equal of course), whose actual or practical yield barely reaches 25%.
What justifies this superiority of "diesel" over the internal combustion engine, with regard to **practical efficiency** , is the very high compression ratio of the so-called combustion diesel engine compared to the internal combustion engine.

To fully understand the extraordinary results obtained with the "G system" on engines, especially diesel engines, it seems essential to me to deepen these stories of efficiency and compression ratio.

Thus, we are content to tell you in the literature, that the higher the compression ratio, the higher the efficiency. Here then is another aspect of thermodynamics which is not easy to highlight in detail, generally, we are satisfied to assert this thing to you without giving you the cause.
I will therefore undertake to describe this phenomenon to you.

But first, here is a little reminder to help you understand the rest.

If you engage a certain amount of work to compress a metal spring, it will give you back when it is released, almost the amount of work invested in its compression since the losses in the form of heat caused during the deformation. of metal are negligible.
Now if you compress any gas in a variable volume vessel like an engine cylinder, the gas will behave like a spring since it is compressible, but unfortunately in reality the work returned when it is released will be less, because there will be a variable part of the work invested which will have been dissipated in the form of heat by the walls of the cylinder. Nevertheless, one still recovers at the time of the relaxation, a large part of the work invested in a variable way according to the speed of the engine and its operating temperature, it is one of the causes of loss of practical efficiency described previously.
Finally, it should be understood that the compression time of an engine is quite expensive in energy, but that it has the enormous advantage of raising the air temperature.**at a very high level,** not by adding calories to it, but by concentrating **the calories initially present in this air** in **a small volume, which allows the fuel to ignite in the case of "diesel".**

It should therefore be understood that:

- The more the air (or the gas) admitted will be compressed into a small volume, the higher its temperature will be despite the losses
- The more this air (or this gas) <u>will have a high temperature before</u> its compression, the more its <u>temperature will be</u> temperature <u>will be high</u> after its volume reduction. These two points must be well understood because this will become very important in the diesel version "Pantone G system".

As you have understood from reading the previous lines, it is the increase in volume of a gas, obtained by its rise in temperature, itself obtained by a supply of calories (combustion of a fuel), which allows this gas by **its expansion** to finally provide us with a certain amount of work.
Thus in an engine, the efficiency is all the higher as the difference between the maximum temperature of the gases at the time of combustion and the minimum temperature at the opening of the exhaust valve is important. What, in fact, must be understood especially for **the pressure difference** between the maximum and the minimum.
But looking at it more closely, one might think that it is only the temperature reached at the end of compression, just before combustion that is important. This temperature being all the higher as the pressure is high (depending on the compression ratio).
This leads us to think that for this reason alone it will be necessary to inject less fuel (calories) if the temperature of the compressed air is already very high (in the case of diesel) to reach the temperature level, i.e. the desired pressure level on the piston.

It is good to understand this kind of thing, but we must not be fooled, **because in reality, what makes the increase in efficiency is not the increase in temperature obtained by the compression, which is not than work taken from the previous engine time, but the fact that it takes <u>less and less calories</u> to raise to a certain temperature the same quantity of any gas as and when it <u>is compressed into a volume more and more restricted</u>, this independently of the normal increase in temperature which follows due to this compression.**

Therefore, the **<u>higher the compression ratio of an engine, the less energy is required to obtain the temperature difference, or the pressure difference mentioned above.</u>**
Thus, for the same work produced, the engine consumes less fuel.
So this is exactly what we have in the case of the diesel engine which has a high compression ratio compared to other types of engine.

After all these explanations on the traditional operation of a conventional diesel, here is what happens for a diesel equipped with a “PANTONE system G” device supplied with air humidified by a “bubbler” system or the like.

We take a diesel engine equipped with a PANTONE reactor (which is in fact in this precise case, only a heat exchanger against the current).
On the intake circuit, the engine still sucks air in the normal way, but in addition it sucks in very largely humidified air by bubbling into a container that we call bubbler. This humidified air is sucked through the reactor and is thus brought to a very high temperature since the reactor is also brought to a very high temperature by the exhaust gases which pass through it.
Thus, the humidified air serves as a means for extracting heat lost from the exhaust gases, in order to reintroduce them into the engine in the form of a mixture of steam and very hot air.

What happens from the moment the engine, the reactor and the bubbler are at normal operating temperature:

- 1-Through the normal air inlet (intake pipe) air at room temperature and a certain quantity of water molecules enter the engine if the weather is wet.

- 2- Through this same air inlet, coming from the reactor, air laden with water vapor enters the cylinder, all at very high temperature, the level of which depends on the quality of the heat exchange from the whole. This mixture can therefore reach 300 or 400 ° and even more if the installation is well designed and it is also variable according to the engine speed.

In fact what goes into the engine chemically and thermally is:

- By the normal route, air composed of nitrogen (80%) and oxygen (20%) at ambient temperature with a few molecules of 'water.
- Through the reactor and the normal route, air (ie nitrogen and oxygen) and water vapor superheated to 300 or 400 °, ready to be broken down into hydrogen and oxygen inside the engine.

The most important thing is that you have to understand that heat energy, stored mainly in the water vapor and recovered free of charge from the exhaust gases, enters the engine.

Seen from this perspective, the engine is a kind of heat pump whose water transformed into vapor can be compared to the refrigerant of a conventional heat pump.

- 3- The admission being completed, the piston goes back up in the cylinder and compresses the assembly as in a normal diesel, but in this case the temperature reached is very high. It is quite possible that despite the significant losses of calories by the walls of the cylinder, this temperature greatly exceeds 2000 ° at the end of compression.
At this temperature level, the conditions are favorable for the decomposition or the dissociation of water vapor into H and O to take place, only a sufficient quantity of calories is lacking to bring the decomposition to its end.

- 4-Thus the piston having reached the top dead center, the temperature being around 3000 °, the fuel (diesel or other) is injected, it ignites spontaneously and thus brings the calories which are lacking to trigger the decomposition of a more or less quantity of water vapor present between the piston and the cylinder head.

- 5- Hardly appeared, the hydrogen and oxygen mixture explodes instantly restoring all the heat energy used for decomposition and which will appear in the reappearance of all of the water entered in the engine, in the form of dry steam at very high temperature therefore at very high pressure.

NOTE: Part of the exhaust gas which is only water vapor could almost be reintroduced directly into the engine if it were not encumbered by the presence of nitrogen in the air and carbon dioxide from the combustion of carbon from fuel.

Now that I have described to you the operation of conventional diesel and the operation of the same engine equipped with the "Pantone G system", it seems to me that the difference is clear to you:
In the operation of the traditional engine, the fuel by burning gives us power. heat which heats all the gases present and which appear and which by relaxing directly give us "work".
In operation with system G, the heat produced by the fuel is used to decompose the water vapor previously produced thanks to the calories recovered from the exhaust gases and it is the oxygen and hydrogen resulting from the decomposition which exploding produce work by the expansion of the vapor that appears and all other gases present.

In the traditional diesel engine, the only purpose of the intake time is to get the maximum amount of air into the cylinder in order to have the sufficient quantity of oxygen necessary for the combustion of a maximum of fuel in order to obtain the maximum amount of air. power at full throttle and for engine displacement. Unfortunately, we admit a large quantity of nitrogen in the cylinder which is harmful in terms of **practical** efficiency..
For the same engine equipped with the G system, we do not admit only air, but, in addition, a large amount **of heat energy taken from the exhaust gases** via the water vapor.

Certainly, we admit less air, that is to say less oxygen, but as we burn less fuel, the business is beneficial since we admit less nitrogen, which causes a small gain on the practical output.

The engine therefore admits superheated steam instead of unnecessary air which becomes a "fuel" after its decomposition.
Note that the heat energy obviously lost during the compression of this vapor by the walls of the cylinder, as for the compression of the air, comes from the exhaust and would have been lost anyway, whereas for the compression of air is energy taken from the work produced, which is not the same thing at all, hence the increase in practical efficiency mentioned above.

At the end of compression, when the piston is at top dead center, depending on the speed, a greater or lesser part of the water vapor is broken down by the combustion of the injected fuel. The immediate explosion of the hydrogen-oxygen mixture which has just appeared, means that the rise in pressure is **instantaneous** when the compression ratio is **at the maximum** , which means that **the practical efficiency approaches considerably the theoretical efficiency** which, in fact , is that of an internal combustion engine which would have the same compression ratio (like that of a diesel).

Our diesel engine, said to be combustion, has become an internal combustion engine with a high compression ratio, therefore with high theoretical efficiency with a reduced deviation from practical efficiency.

It should be understood that if the theoretical efficiency of the conventional diesel engine does not have the theoretical efficiency that it should have because of its high compression ratio, it is because the release of the heat energy contained in the fuel does not is not done instantly and entirely when the compression ratio is at the maximum (i.e. at the maximum theoretical efficiency), because due to the prolonged injection, the release of calories is done gradually while the piston goes down, ie while

the compression ratio decreases which results in a simultaneous collapse of the theoretical efficiency (and obviously of the practical efficiency).

After the explosion of the H and O mixture, the vapor produced is added to that which may not have been decomposed.

In fact, in conventional diesel, there is only a small part of the fuel that burns when the compression ratio is at its maximum, that is to say while **the efficiency is at the maximum** : this is the **serious drawback of the diesel engine** which brings its practical output hardly above the internal combustion engine.
In the engine with the G system, this disadvantage disappears, and at this level alone we have a considerable saving in fuel consumption which is added to the other reasons. It can be reduced by more than half depending on the compression ratio.

After the explosion of the mixture of H and O, the vapor produced is added to that which would not have been decomposed during the combustion of the fuel and the expansion of this as well as all the gaseous products present. produce "work" by action on the piston.

The engine behaves in part like a high-efficiency steam engine, the heat source of which, the boiler, is inside the cylinder and part of the lost energy of which is recycled as by the under draft in the steam installations.

Now we can conclude that a "diesel engine equipped with the PANTONE" system G "device has the following advantages over conventional diesel:

1- Increase in practical efficiency by reducing the difference between it and the theoretical efficiency of the makes the heat losses during the compression time cost less labor.

2- Increase in theoretical efficiency due to the fact that the combustion engine operates more and more as an internal combustion engine when it is asked to provide more and more power. At the same time there is a considerable increase in practical efficiency.

3- The disadvantage of poor air filling of the cylinder mentioned above is no longer one for practical efficiency.

4- There remains only the disadvantage of the truncated trigger (see my page 16), but an improvement of the problem is technically difficult to envisage and moreover irrelevant in the present case.

We immediately understand that, depending on the importance of improvements in theoretical and practical efficiency, the consumption of an engine can collapse by more than half without taking into account the recovery of part of the heat energy of the engine. 'exhaust. If we add the recovery of this energy (see my page 15), consumption still collapses in a variable way depending on the engine speed.
This explains, without mystery, that depending on the quality of the construction and the type of diesel thus equipped, consumption three times lower than normal can be observed.

NOTE: It would be interesting to suppose that despite the excellent results observed during the operation of the diesel thus equipped, it would be judicious to "fine-tune" the adjustment of the advance to the fuel injection so that the explosion of the mixture O and H does not occur until the piston reaches top dead center. Since fuel injection is primarily intended, once the engine is warm, to decompose water vapor, it would probably be preferable to shorten the duration of this injection.
I also think that it is not necessary to admit too much steam into the engine, but to admit it at the highest possible temperature.

There are many things to study about this engine which at its temperature no longer works as a diesel engine but as an internal combustion engine with a high compression ratio and therefore **high efficiency.**

In short, a lot of research work remains to be done on this device and must be undertaken by people with the appropriate technical means.

Although it is spectacular and “media” to say that an engine thus equipped is a water engine, it must be admitted that it is not. In the case of an engine equipped with a "Pantone G system", it is always a hydrocarbon engine, only the means of deriving maximum benefit from it, that is to say to obtain a maximum of fuel. efficiency of this engine, goes through the use of water which fundamentally changes its operation which is ultimately only possible with the energy of the hydrocarbon used.

Indeed, it is obvious that everything that happens to the water involved in this matter, all the real or supposed phenomena, such as vaporization, decomposition, dissociation, the appearance of plasma sometimes mentioned, etc., in n 'any part of the engine or the Pantone system, are only possible by the energy released by the fuel used in the installation, because this fuel carries the only energy that enters the engine, because unlike the water it is like a "taut spring" ready to relax with a tiny "matchstick crack".

Please also note that at the end of a cycle not a drop of used water is wasted. If we take the trouble to distill the exhaust gases, all of the water used in the system will be recovered as well as that coming from the humidity of the air and that coming from the hydrogen part of the hydrocarbon. In fact, the engine does not consume a drop of water and gives us back more than we have provided.

On the other hand, the hydrocarbon has completely disappeared and all that remains of it, apart from the water vapor, is a gas that is energy-neutral (CO_2).

In our case, water is only a "relaxed spring" which is only capable of giving us back the energy that we have given it.

Water is only the result of the combustion of hydrogen with its oxidizer oxygen, and as one of my friends says so well, water is only "ash".

Despite its resemblance to the original PANTONE system in which water and hydrocarbon pass together, undergoing radical transformations, the "G system" is quite different in terms of engine operation where these transformations take place in the engine. this.

I think this is my last "Page", I leave it to others to express themselves on this subject, to go into every detail and to make the corrections to any errors I might have made. DAVID Michel

"Xav tractor!"

Bernadette and Jean: We warmly thank our friend Michel David who spent a lot of time writing his "Pages", nearly a hundred in all, where many experimenters of Pantone or "Pantone" assemblies found material. on reflection.

It is evident that as the very first experimenter of a Pantone reactor in France, he was in the best position to provide his thoughts and improvements to the system.

Now the results are known to thousands of people, and the early thoughts of the 2000s and 2001s may seem straightforward to some. But there are still a lot of tips he gave (some of which to prevent patents as "untimely" as useless from being filed) which have unfortunately never been tried (to our knowledge), and this is a shame as for example rotary distributors.

Failing to be able to experiment everything himself (for lack of time) he wanted to give his way of seeing things, and provided his many advices so that the experimenters who started to make Pantone assemblies can go further. . From "young" engineers, many correspondents told us that they learned a lot thanks to its texts and diagrams. And we even remember that at the time when Pantone was leading training courses, he had Michel David's pages translated for his students, that says it all ...

His most sincere wish, since we know that this highly interesting system continues to prove its undeniable effectiveness thanks to the hundreds (even thousands) of positive replications, is that institutes, schools, teachers, scientists, professionals in laboratories well equipped can fine-tune over and over again ... a system that may sound too simple but is actually great.

Page 1 Page 2 Page 3 Page 4 Page 5 Page 6 Page 7 Page 8
Page 9 Page 10 Page 11 Page 12 Page 13 Page 14 Page 15 Page 16

Menu **Summary** **Home**